Uncovering spatial distribution of biomolecule functionalised PAcrAm™-g-PMOXA coated surface using DNA-PAINT

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Abstract

It is important for biosensors to have a low-fouling layer to reduce non-specific interactions, and a high-density, homogeneous bio-functionalised surface to ensure reliable and reproducible results. While PLL-g-PEG is a well-researched polymer, recent studies demonstrated that it is not ideal for biomedical applications. PLL-g-PEG has a limited storage lifetime due to the repetitive oxygen atom in the PEG side chain, and the surface interactions are not as strong as other molecular bonds. A promising alternative, PAcrAm™-g-PMOXA, addresses these limitations with a stronger stability for increased storage lifetime and a stronger surface adhesion. While many properties of PAcrAm™-g-PMOXA have been researched, the spatial properties of the biomolecule functionalised polymer coated surface have not been investigated. With the use of DNA-PAINT, the surface properties were measured at a single-molecule resolution. Using different data analysis techniques, the surface density and spatial distribution have been determined. This study revealed several things. The formation of the low-fouling layer does not happen instantaneously after surface immobilisation. The polymer can rearrange itself to form an equilibrium configuration before the covalent bonds are formed. The observed number of affinity binders per polymer decreases with increasing DBCO-ssDNA incubation concentration. Lastly, the ssDNA-functionalised PAcrAm™-g-PMOXA showed a random distribution.

1 Introduction

Non-fouling surface treatments prevent adhesion of biomolecules such as proteins, cells, and micro-organisms onto the surfaces [1]. The application of these non-fouling surfaces is important for a wide range of fields, from drug-delivery systems [2] and biosensing [3] to water supply [4] and ship hull coatings [5]. It is important for biosensors to reduce non-specific interactions during a measurement. The non-specific interactions can interfere during the measurements, decreasing the precision of the measured signal [6]. The most promising low-fouling behaviour for surface treatments has been achieved through the use of uncharged hydrophilic or zwitterionic polymers [7]. The low-fouling properties of polymers are achieved by having hydrogen-bond acceptors, polar functional groups, the absence of net charge, and no hydrogen-bond donor groups [1], [8]. In addition to a low-fouling layer, obtaining a high binder density and a homogeneous surface is crucial for biosensors. Many biomolecules are in the range of pico- and nanomolar, and a high surface binder density ensures accurate signal during a measurement [9]. Typical surface affinity binders range from 10^2 to 10^5 molecules per square micrometer. A high-density, homogeneous surface with a low-fouling layer is essential for optimal sensor performance.

A well-researched low-fouling surface treated polymer for biomedical applications is poly(l-lysine)-graft-poly(ethylene glycol) (PLL-g-PEG) [10], [11]. The PLL backbone is positively charged and binds onto negatively charged surfaces via electrostatic interactions [12]. The PEG side chains have a neutral net charge, are a highly hydrophilic polymer with polar moieties, show hydrogen-bond acceptors, and have no hydrogen-bond donors. This results in strong interactions with water molecules via hydrogen bonds. The PEG side chains adopt a brush-like conformation and form a strong hydration layer that prevents protein adsorption entropically and sterically [13], [14]. Bio-functionalised PLL-g-PEG coated surfaces have shown successful use in biosensors. For instance, Buskermolen et al. [15] demonstrated a continuous biomolecular sensing method based on the diffusion of bio-functionalised particles that was able to detect in the picomolar range. Furthermore, biotinylated PLL-g-PEG has shown promising use for immunoassays. Huang et al. [16] showed specific interactions of linkage proteins such as streptavidin and avidin, while maintaining a high resistance against non-specific interactions from human blood serum. Moreover, Tan et al. [17] have described the spatial molecular properties of ssDNA-functionalised PLL-g-PEG molecules. The spatial molecular heterogeneity of ssDNA-functionalised PLL-g-PEG molecule varies depending on the conditions of the bio-functionalisation. Increasing the bio-conjugation time resulted in a more dispersed molecular distribution, whereas additional incubation steps led to a decreased molecular dispersity.

In recent observations, the PEG side chains are far from ideal for biomedical applications [18]. Even though PEG is quite stable in physiological environments, it is, however, oxidative in aqueous environments. The repetitive oxygen atom in its structure can degrade and has the potential to increase the production of anti-PEG antibodies in vivo [19]–[21]. As a result, the use of PEG is hindered by a limited storage lifetime [22]. In addition, the PLL backbone only interacts electrostatically to a negatively charged surface. Electrostatic interactions are not as strong of a bond compared to other molecular bonds that can be formed. With these considerations, interest in alternative anti-fouling polymers increased to overcome these limitations [23].

A promising alternative is poly(acryl-amide)-graft-(poly(2-methyl-2-oxazoline)$_{50}$, 1,6-hexanediamine, 3-amino propyldimethylsilanol) (PAcrAm™-g-PMOXA$_{50}$, NH2, Si), which will be denoted as PAcrAm™-g-PMOXA hereafter. The PAcrAm™ backbone binds electrostatically with the amine side chain and covalently with the silane side chain onto the surface. Due to the addition of the covalent bond, PAcrAm™ is more stable on the surface compared to a PLL backbone [1], [7], [13]. The low-fouling layer is created by the zwitterionic PMOXA side chain. Compared to the hydrophilic PEG polymer, the hydration layer created by zwitterionic polymers is denser and thicker. As the barrier is stronger, biomolecules have more difficulty penetrating this barrier, making it adhere less to the surface [24]. In addition to a stronger hydration layer, the PMOXA side chain does not contain oxygen atoms, preventing oxygen degradation and resulting in a longer storage lifetime [25].

Different properties of PAcrAm™-g-PMOXA have been researched and compared to PLL-g-PEG [7], [20], [25]. However, the spatial distribution of the biomolecule functionalised polymer has not been investigated, even though it is important for interactions in the environment and ensuring reproducibility [26]. Surface characterisation techniques such as quartz crystal microbalance [27], surface plasmon resonance [28], and fluorescence microscopy can determine average surface densities, but lack the spatial resolution to characterise surface distributions on a molecular level. To overcome the spatial resolution, a subgroup of super-resolution microscopy called single-molecule localisation microscopy (SMLM) has been used. SMLM achieves super-resolution by temporally separating blinking events to eliminate overlapping point spread functions (PSFs) [29]. The specific technique used in this work is DNA point accumulation in nanoscale topography (DNA-PAINT) [30]. Single-molecule resolution is achieved by the transient binding kinetics of DNA, using a single-stranded DNA (ssDNA) docking strand on the molecule of interest and a complementary dye-labelled ssDNA imager strand in solution. Compared to other SMLM techniques, DNA-PAINT is not limited by photobleaching and is dependent on DNA binding events to create '*apparent*' blinking [31]. The kinetics of DNA binding allows for control over the blinking events for specific applications.

This study focuses on characterising spatial heterogeneity of bio-functionalised PAcrAm™-g-PMOXA using DNA-PAINT. The surface densities are determined by using three analysis methods. One to determine the ssDNA-functionalised polymer density, one to determine the average number of affinity binders per polymer, and one to determine the affinity binder density. Furthermore, the spatial distribution of the ssDNA-functionalised polymer has been determined. The effects of varying PBS incubation time, varying DBCO-ssDNA incubation time, and the combination of both have been investigated. The study indicated that an additional PBS incubation helps the polymer to form a homogeneous, low-fouling surface. Increasing the DBCO-ssDNA incubation time introduces a higher average number of affinity binders per polymer. Increasing the ssDNA incubation concentration does not necessarily mean an increase in the observed average number of affinity binders per polymer, which may be caused by accessibility issues due to the high-density surface.

2 Experimental Setup

2.1 Materials

Glass coverslips (22 × 40 mm, thickness #1, Menzel-Gläser) were obtained from VWR. Custom-made flow cell stickers with an approximate internal volume of 20 μL were obtained from Grace Biolabs (USA). Poly(acryl-amide)-graft-(poly(2-methyl-2-oxazoline)$_{50}$, 1,6-hexanediamine, 3-amino propyldimethylethoxysilane) (PAcrAm™-g-PMOXA$_{50}$, NH_2, Si) and Poly(acryl-amide)-graft-(poly(2-methyl-2-oxazoline)-N_3, 1,6-hexanediamine, 3-amino propyldimethylethoxysilane) (PAcrAm™-g-PMOXA-N3, NH_2, Si) were purchased from SuSoS (Switzerland). Fluorescent nanoparticles (0.2 μm, yellow-green, Molecular Probes) were used as fiducial markers. PBS tablets, NaCl, and Mg_2Cl were purchased from Sigma-Aldrich, and Tween 20, EDTA, and Tris-HCl were purchased from Merck Life Science. The ssDNA oligonucleotides (standard desalting and HPLC purification for chemically modified DNA) were purchased from IDT (Integrated DNA Technologies). The ssDNA sequences used are shown in Table 1. Sequence 1 is the dibenzocyclooctyne (DBCO) strand, which clicks onto the azide-functionalised polymer via Strain-promoted azide-alkyne cycloaddition (SPAAC) click chemistry. This sequence is pre-hybridised with either sequence 2 or 3, the complementary and non-complementary ssDNA docking strand, respectively. The pre-hybridised DNA will be denoted as DBCO-ssDNA hereafter. Sequence 4 is the imager strand. The imager strand has the fluorophore ATTO 647N already attached to it with an absorption and emission wavelength of 646 nm and 664 nm, respectively. PBS buffer was prepared by dissolving 1 tablet of PBS in 200 mL of Milli-Q water. 1 M of NaCl dissolved in PBS was used as the high-salt (HS) buffer in this study. The imaging buffer (Buffer B) consists of 10 mM Mg_2Cl, 5 mM Tris-HCL, 1 mM EDTA and 0.05% Tween 20.

Table 1: ssDNA sequences

#	5' end	sequence	3' end
1	N/A	CGATTCGAGAACGTGACTGCTTTTT	DBCO
2	N/A	GCAGTCACGTTCTCGAATCGAACATTATTACA	N/A
3	N/A	GCAGTCACGTTCTCGAATCGAAGTAATAATG	N/A
4	ATTO 647N	TTGTAATAATG	N/A

2.2 Sample Preparation

A 10 μM stock solution of pre-hybridised DBCO-ssDNA was used as the docking strands. The stock was prepared by hybridising a 4:1 molar ratio of binder ssDNA to ssDNA labelled DBCO, both with a concentration of 100 μM, in a HS buffer on a rotating fin for at least 2.5 hours. The coverslips were washed with isopropanol for ten minutes in a sonic bath, followed by MilliQ for ten minutes, and dried with a nitrogen stream afterwards. The surface of the coverslips was oxidised with plasma oxygen, a flow cell sticker was applied onto the coverslip, and the flow cell was filled with the polymer solution and incubated for three hours in a humidity chamber. The polymer solution consists of a 1% v/v PAcrAm™-g-PMOXA-N3/PAcrAm™-g-PMOXA ratio, made from the stock solution of 10 mg/mL concentration. The polymer solution was diluted with MilliQ to a final combined concentration of 0.1 mg/mL. After three hours, the polymer mixture was replaced with PBS for an extra incubation step before the DBCO-ssDNA solution. If an extra PBS incubation step was not needed for the experiment, the polymer solution was replaced with a 2 μM DBCO-ssDNA solution, unless stated otherwise. In this work, the PBS incubation duration and DBCO-ssDNA incubation duration were varied. Before imaging, the stock solution of the fiducial markers was sonicated for five minutes. The sonicated stock solution was diluted 10,000 times with PBS and sonicated again for five minutes before use. To prepare the samples for imaging, the samples were washed with 100 μL PBS and the buffer was replaced with the fiducial marker solution. The fiducial markers were allowed to sediment for five minutes. The unsedimented particles were flushed away with 100 μL Buffer B. Lastly, the imager solution, diluted with Buffer B to 25 pM, was added.

2.3 DNA-PAINT Imaging

DNA-PAINT was performed using the Oxford Nanoimager (ONI) with the total internal reflection fluorescence (TIRF) setup. The ONI has a 100× 1.4 NA oil submersion objective. Two lasers with a wavelength of 532 and 640 nm were used to obtain the images with a power of 0.1 and 33.0 mW, respectively. The measurement duration was an hour with a frame rate of 10 Hz. The fluorescence signal passed through a beam splitter into a green and red channel, where the fiducial markers were visible in the green channel only, and the binding events were only visible in the red channel. The raw data obtained from the ONI were post-processed using ThunderSTORM [32], an open-source ImageJ [33] plug-in, to obtain spatial and temporal information, and other parameters of the detected fluorescence. The obtained information from ThunderSTORM was further analysed to correct for drift, filter the data, and determine the molecular density and distribution.

2.4 Data Analysis

Three different analysis methods were used to determine the surface densities, and one analysis method to determine the spatial distribution of the ssDNA-functionalised polymer. The analysis methods followed the research of Tan et al. [17]. The methods are briefly explained here; a more detailed explanation will be given in Section 3.1. The first method, Direct Counting (DC), determines the polymer density. This approach uses a mean-shift clustering algorithm to determine the observed ssDNA-functionalised polymer density. In a high-density system, not all ssDNA-functionalised polymer molecules will be probed due to the limited measurement duration. The algorithm classifies binding events within a bandwidth of twice the maximum localisation uncertainty to one polymer. The second method, Kinetic Counting (KC), determines the average number of affinity binders per polymer. This method, based on quantitative-PAINT [34], uses the observed unbound state lifetime to determine the average number of affinity binders per polymer. DC and KC are limited by the measurement duration and the imager concentration. These limitations cause an underestimation of the true number of ssDNA-functionalised polymers, total ssDNA molecules, and mean unbound state lifetime. This causes an overestimation of the average number of affinity binders per polymer. Monte-Carlo simulations were used to compensate for the molecular undersampling. Compensation for Lifetime Undersampling (CLiU) was used to determine a more accurate average number of affinity binders per polymer. The third method, Compensation for Binder Undersampling (CBiU), was used to compensate for the observed number of binding events to determine the binder density. CLiU and CBiU simulated the binding events of the docking and imager strands, with the parameters shown in Table 2.

Table 2: Input parameters for the Monte-Carlo simulations

c_{imager}	$[M]$	$25 \cdot 10^{-12}$
k_{off}	$[s^{-1}]$	1
k_{on}	$[(M \cdot s)^{-1}]$	10^6
Frame rate	[Hz]	10
t	$[s]$	3600

The ssDNA-functionalised polymer density was determined using DC analysis, which provides insight into the observed ssDNA-functionalised polymer density. Additionally, CLiU analysis was used to visualise the average number of affinity binders per polymer, while CBiU analysis was used to determine the affinity binder density. These two methods compensate for the experimental limitations, leading to a more accurate estimation.

Lastly, the polymer distribution was determined using a statistical method based on the nearest-neighbour distance (ndd) proposed by P. J. Clark and F. C. Evans [35]. The observed distribution is compared with the spatial distribution under the complete spatial randomness (CSR) hypothesis. Rejecting the CSR hypothesis indicates that the ssDNA-functionalised polymer-coated surface is not randomly distributed.

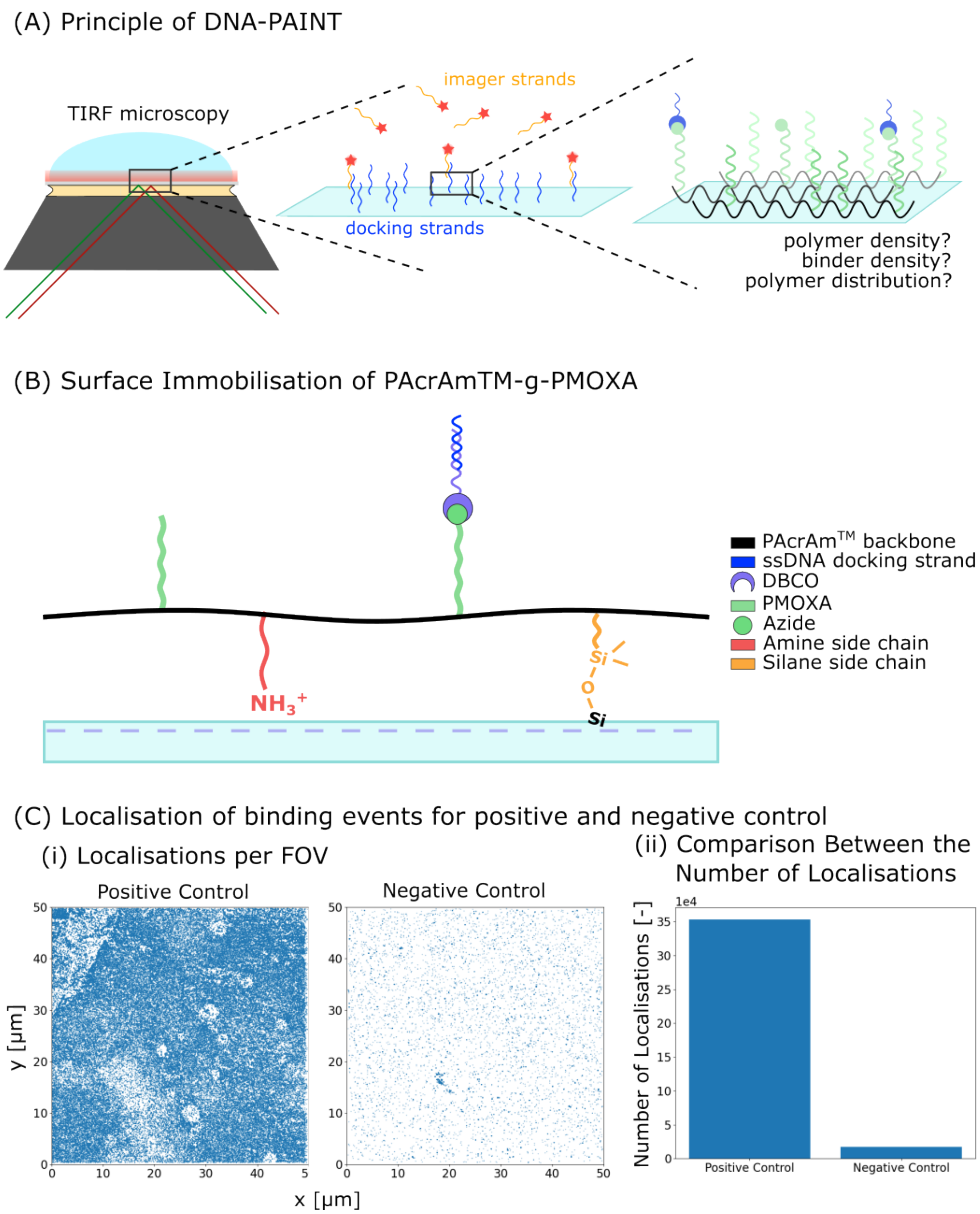

Figure 1: (A) DNA-PAINT is performed using the TIRF setup to detect binding events between imager strands and docking strands close to the surface. (B) PAcrAm™-g-PMOXA binds to the surface covalently with the silane side chain, and has electrostatic interactions on the surface with the amine side chain. Azide-functionalised PMOXA side chains can react with DBCO-ssDNA via SPAAC click reaction. (C) The coverslips were incubated for 3 hours with the polymer solution, followed by a 3-day incubation with 2 DBCO-ssDNA concentration. The positive control (10 bp complementarity between docking and imager strands) shows a lot of detections, while the negative control (9 bp non-complementarity between docking and imager strands) shows few detections.

3 Results and Discussion

3.1 DNA-PAINT analysis of PAcrAm™-g-PMOXA functionalisation

The experiments are performed using DNA-PAINT on the TIRF setup, as shown in Figure 1A. The pre-hybridised DBCO-ssDNA were conjugated via SPAAC click chemistry on the azide-functionalised PMOXA side chains to function as the docking strands, as illustrated in Figure 1B. The DBCO-ssDNA docking strands were either complementary or non-complementary to the imager strands. The positive control in Figure 1C has a 10 bp overlap between the docking and imager strands. A high number of localisations have been detected that are specific between the docking and imager strands. The negative control in Figure 1C has no complementarity between the docking and imager strands, with the docking strand being 9 bp long. A low number of localisations have been observed, indicating non-specific interactions. The localisation data from ThunderSTORM were first corrected for drift. The fiducial markers were used during the experiment to measure the drift. Following, the data were filtered to exclude falsified identified localisations. The filtering was based on three criteria: the background offset, the width of the PSF, and the localisation uncertainty. The background offset corrects for the intensity difference between the background of the localisation and the camera baseline. If the width of the PSF of the localisation is unusually large or small, it was also disregarded. The localisation uncertainty is the uncertainty of the position of the identified localisation. Uncertainties larger than twice the mean size of the localisations were disregarded. Following the filtering procedure, smaller 5×5 μm^2 regions of interest (ROIs) were taken from the whole field of view (FOV) that has an area of 50×50 μm^2. From the 100 available ROIs, the edges were discarded to avoid edge effects, and the remaining 64 ROIs were used for further analysis.

Three different analysis methods were used to determine the surface densities. The first method, Direct Counting (DC), determines the ssDNA-functionalised polymer surface density. DC uses a mean-shift clustering algorithm to classify binding events within a bandwidth as one cluster to form a localisation cloud, where each localisation cloud represents the location of one polymer, as shown in Figure 2B. The chosen bandwidth is roughly 30 nm, which is twice the maximum localisation uncertainty. DC only determines the number of ssDNA-functionalised polymers on the surface. The second method determines the average number of affinity binders per polymer per ROI. Kinetic Counting (KC), a method based on quantitative-PAINT [34], used the temporal information of each localisation cloud; from this, the experimental association rates were computed based on the unbound state lifetime per ROI, seen in Figure 2C. The number of ssDNA affinity binders per ROI can be computed using Equation 1.

$$k_{on}^{exp} = k_{on} \cdot c_{img} \cdot \bar{n}_{ssDNA}^{exp} \tag{1}$$

k_{on}^{exp} denotes the experimental association rate, k_{on} the origami-equivalent association rate, c_{img} the imaging concentration, and $\bar{n}_{ssDNA}^{exp}$ the average number of ssDNA affinity binders per polymer. The origami-equivalent association rate (k_{on}) was assumed to be $10^6 (M \cdot s)^{-1}$, as many origami-equivalent studies have quantified this value for well-defined docking strands. However, the experimental association rate is limited by the measurement duration of one hour and lacks information for unbound state lifetimes longer than the experiment duration. Monte-Carlo simulations were used to compensate for the lifetime undersampling effect, named Compensation for Lifetime Undersampling (CLiU). The relation between the average number of ssDNA affinity binders on the polymer via KC and CLiU is shown in Equation 2.

$$\bar{n}_{ssDNA}^{exp} = a \cdot \bar{n}_{ssDNA}^{CLiU} + b \tag{2}$$

a and b denote the slope and intercept obtained with Monte-Carlo simulations, which are 0.8 and 13.0, respectively. $\bar{n}_{ssDNA}^{CLiU}$ is the compensated average number of ssDNA affinity binders per polymer.

The number of binding events is also limited by the experimental duration, which influences the number of ssDNA affinity binders detected on the surface. The binder density was compensated with a Monte-Carlo simulation, named Compensation for Binder Undersampling (CBiU), which is the third analysis method. The simulation simulated time traces for each binder with varying the binder density with the association rate ($k_a = k_{on} \cdot c_{imager}$) and the dissociation rate ($k_d = k_{off}$). The dissociation rate is experimentally determined to be roughly 1 s^{-1}. The probability of the binder and imager strands being bound can be determined by Equation 3.

$$P(\text{bound}) = \frac{c_{imager}}{c_{binders} + K_d}; \quad \text{Where: } K_d = \frac{k_{off}}{k_{on}} \tag{3}$$

This equation is only valid if the imager concentration is much smaller than the binder concentration, and the binder concentration is much smaller than the dissociation constant K_d ($K_d \gg c_{binders} \gg c_{imager}$). These values are estimated to be in the range of μM, nM, and pM, respectively. The relation between the ssDNA affinity binder density and the number of binding events is shown in Figure 2D.

The distribution of the ssDNA-functionalised polymer coated surface has been examined by a statistical method based on the nearest-neighbour distance (nnd) proposed by P. J. Clark and F. C. Evans [35]. Comparing the observed data with a reference distribution based on complete spatial randomness (CSR), the Clark-Evans test determines whether the observed data deviates from the CSR hypothesis to a more clustered or a more dispersed spatial distribution. From the experimental data set, N subsets of size m are chosen to determine individual distribution scores to ensure reliability, as shown in Equation 4.

$$z_m^k = \frac{\overline{D_m}^k - \hat{\mu}}{\hat{\sigma}} \quad \text{Where: } \hat{\mu} = \frac{1}{2\sqrt{\lambda}} \quad ; \sigma = \sqrt{\frac{4-\pi}{4m\lambda\pi}} \tag{4}$$

Where $\overline{D_m}^k$ denotes the mean nnd for subset k (where $k \in [1, N]$), λ is the ssDNA-functionalised polymer density obtained through DC, and $\hat{\mu}$ and σ are the mean and standard deviation of the nnd for CSR, respectively. The distribution score that is used to test against the CSR hypothesis is the average of all the distribution scores obtained with Equation 4, shown in Equation 5.

$$\overline{z_m} = \frac{\sum_{k=1}^{N} z_m^k}{N} \tag{5}$$

With a 5% significance, $|z_{0.05}| = 1.65$, scores outside this range reject the CSR hypothesis, and the polymer spatial distribution is not random. It is either more clustered ($\overline{z_m} < -z_{0.05}$) or more dispersed ($\overline{z_m} > +z_{0.05}$).

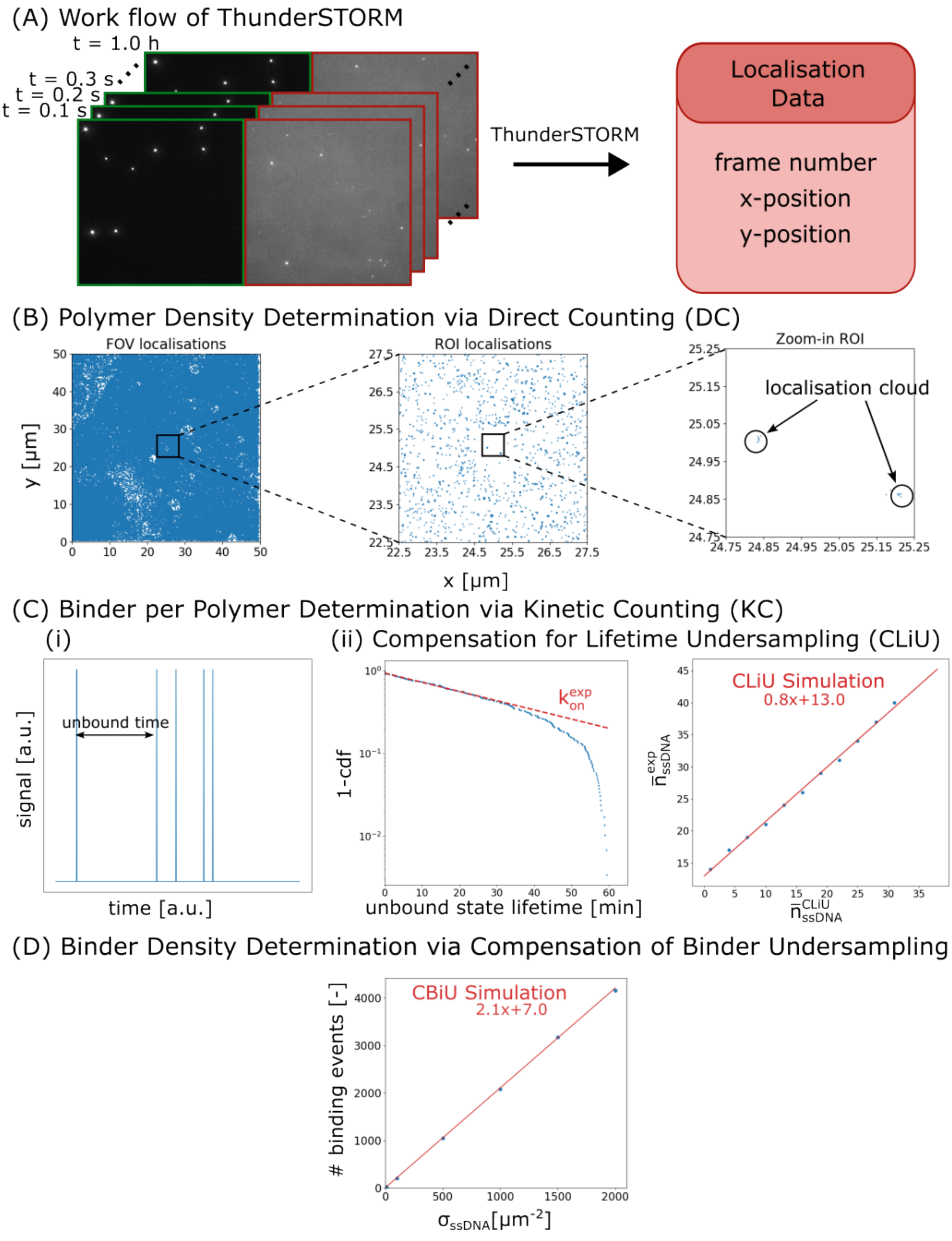

Figure 2: (A) Experimental data is analysed with ThunderSTORM to obtain information about the detected localisations. (B) Direct Counting (DC): The detection of ssDNA-functionalised polymers (localisation clouds) with the use of a mean-shift clustering algorithm to directly count the number of localisation clouds. (C) Kinetic Counting (KC): The unbound state lifetimes were used to obtain the experimental association rates to determine the average number of binders per polymer per ROI. Compensation of Lifetime Undersampling (CLiU): A correlation between the experimental average number of affinity binders per polymer per ROI and the CLiU average number of affinity binders per polymer has been established to compensate for the experimental limitation of unbound state lifetimes longer than the measurement duration. (D) Compensation Binder Undersampling (CBiU): A correlation between the number of binding events and the binder density has been established to compensate for the number of binding events limited by the measurement duration.

3.2 Effect of Varying PBS Incubation Time

The surface immobilisation of the polymer involves both electrostatic interactions and covalent bonds. As stated by SuSoS [36], the covalent bonds form more slowly than the electrostatic interactions. Investigating the PBS incubation will reveal if the formation of the low-fouling layer depends on the covalent bonds or if electrostatic interactions are sufficient for its formation. For these experiments, PAcrAm™-g-PMOXA was used with no azide-functionalised PMOXA side chains; therefore, no docking strands are conjugation onto the polymer. Due to the absence of the docking strands, the detected signals are non-specific interactions either from the interactions between the imager strands and the polymer, or from background noise, such as the glass coverslip.

Figure 3B(i) shows a decrease in the number of localisations with increasing incubation time in PBS, where it stabilised to a minimum after 4 days of incubation. A high number of non-specific interactions has been detected with no PBS incubation. This implies that interactions between polymer and imager strands are detected. An additional incubation step in PBS is required to form a stable low-fouling layer, which reduces the non-specific interactions between the polymer and the imager strands. A drastic decrease in the number of localisations has been observed with the repeats of no additional PBS incubation step, shown as the blue bars in Figure 3B(ii). The number of localisations of the repeats was still higher compared to the 4-day incubation, which indicates that the low-fouling layer is not stable or homogeneous. The green bars in Figure 3B(ii) illustrate the repeats of the 4-day incubation in PBS. The two samples show consistency in the number of localisations. This indicates the reliability and reproducibility of the polymer surface after a 4-day incubation in PBS. The low-fouling layer has been established, and the polymer is in its equilibrium configuration.

From the results, it is observed that the low-fouling layer has not been fully formed before a 4-day incubation in PBS. A hypothesis is that the polymer immobilisation leads to surface heterogeneity, characterised by patches of low-fouling areas. As illustrated in Figure 3C(i), this could be due to incomplete coverage on the surface, unbound polymer tail-ends, or the presence of loops within the polymer backbone. Increasing the incubation time in PBS reduces the number of localisations. The polymer is capable of rearranging to form a homogeneous low-fouling layer, as shown in Figure 3C(ii). This rearrangement can only occur if the covalent bonds have not yet formed. This observation suggests two possibilities: first, the low-fouling layer becomes stable after the formation of the covalent bonds. Second, sufficient covalent bonds are formed after 4 days of PBS incubation, resulting in a homogeneous, low-fouling layer.

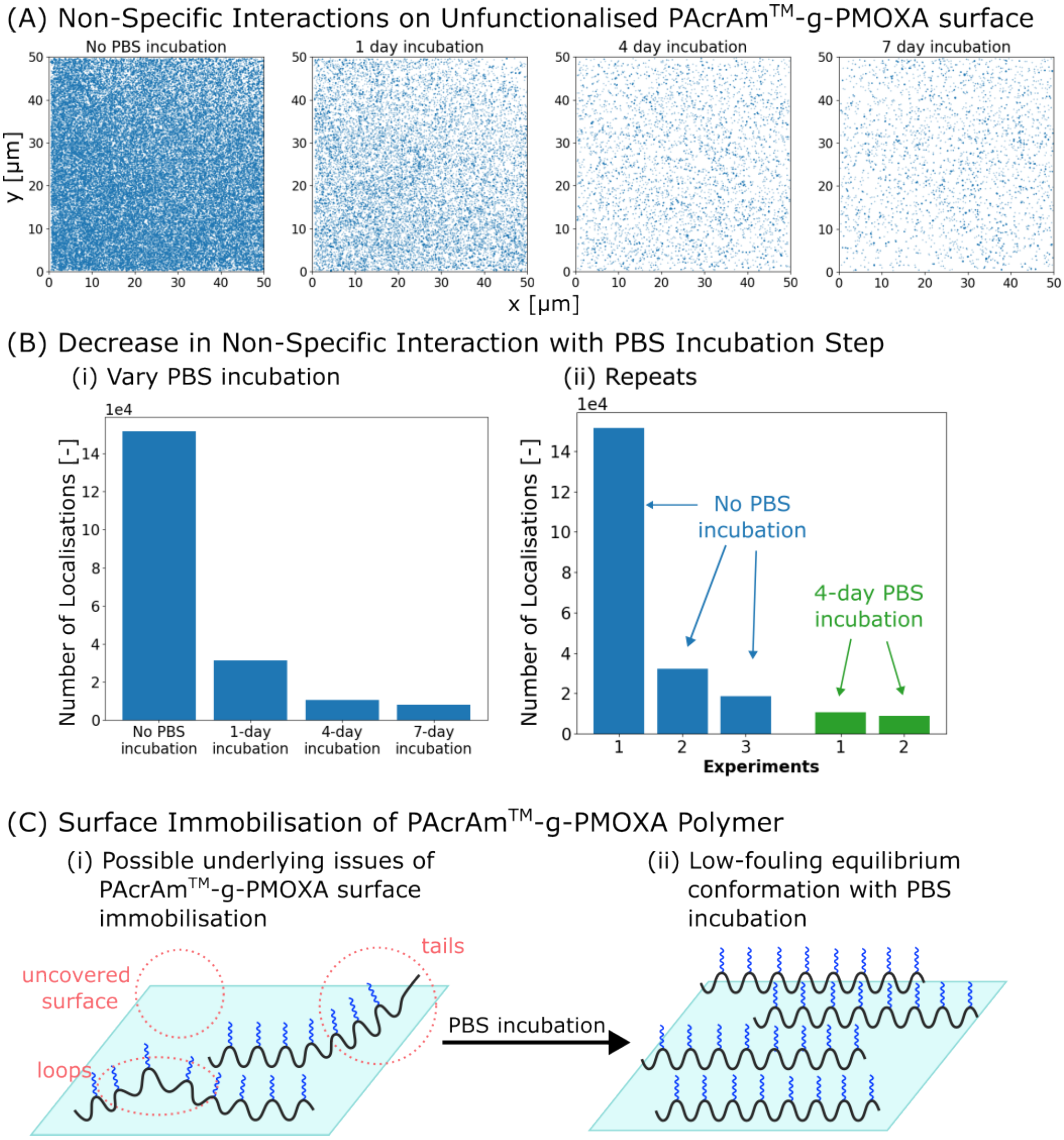

Figure 3: (A) The localisations per FOV of the non-specific interactions decrease with increasing PBS incubation duration. (B) (i) Comparison of the number of localisations between the different incubation times, where a decrease in the number of localisations is detected with increasing incubation time. (ii) Repeats of no PBS incubation and 4-day incubation were done, where a decrease was observed with no PBS incubation. The 4-day incubation shows a reproducible value. (C) (i) Underlying issues of the surface immobilisation that might cause the high number of localisations. (ii) Incubating in PBS results in a more homogeneous coated surface with a stable, low-fouling layer.

3.3 Effect of Varying DBCO-ssDNA Incubation Time

The SPAAC-click reaction is generally slow compared to other click-chemistry reactions, with a second-order rate constant between 1-60 $(M \cdot s)^{-1}$ [37]. A longer incubation time with the DBCO-ssDNA concentration should increase the number of affinity binders on the surface. These experiments did not include an additional PBS incubation step.

Figure 4A shows an increase in the number of localisations with increasing incubation time and Figure 4B shows the increase in ssDNA-functionalised polymer density (4B(i)) and affinity binder density (4B(ii)) with increasing incubation time. This indicates an increase in conjugated ssDNA onto the azide-functionalised polymer with increasing incubation time. The average number of ssDNA affinity binders per polymer also shows an increase with increasing incubation time, as shown in Figure 4C. The observed average number of ssDNA affinity binders per polymer ranges from 2.1 ± 1.10 binders per polymer to 4.1 ± 0.9 binders per polymer. Each polymer has, on average, more ssDNA affinity binders.

The experimental values are lower compared to the theoretical calculations (see Appendix A). One of the reasons for the distinct difference lies in the assumptions made for the theoretical calculations. The calculations regarding the geometry of the polymer assume a well-extended polymer on the surface. In experimental settings, polymers are not always well extended and can behave differently. In addition, the used k_{on} value is for well-defined DNA-origami structures with well accessible docking strands. Experimentally, accessibility issues arise with a high-density surface. Not all docking strands could be accessed, resulting in a lower association rate.

Notably, the 1-day incubation in the DBCO-ssDNA concentration deviates from the general trend. It yielded a higher number of localisations compared to the 3-day incubation and a higher surface density compared to the 3-day and 4-day incubation. The reproducibility of the 1-day incubation showed a similar number of localisations, shown in Appendix B, Figure A. This implies that it is not out of the ordinary. A hypothesis is that an interplay between the polymer and the imager strands is present. From section 3.2, non-specific interactions were still detected after 1-day incubation in PBS. Assuming that 1-day incubation in the DBCO-ssDNA concentration has a similar effect on the polymer as 1-day incubation in PBS, it could be that the polymer did not have enough time to form a homogeneous low-fouling layer. Interplay between the imager strands and the polymer is possible on the patches where the low-fouling layer has not been formed yet. Given these non-specific interactions, the densities shown in Figure 4B cannot be accurately interpreted as true densities and are included solely for the representation of the figure. Another source of possible non-specific interactions might arise from the unconjugated azides. In a shorter incubation time, fewer DBCO-ssDNA can click onto the azide-functionalised polymer, leaving unconjugated azides. This results in non-specific interactions between the imager strands and the unconjugated azides.

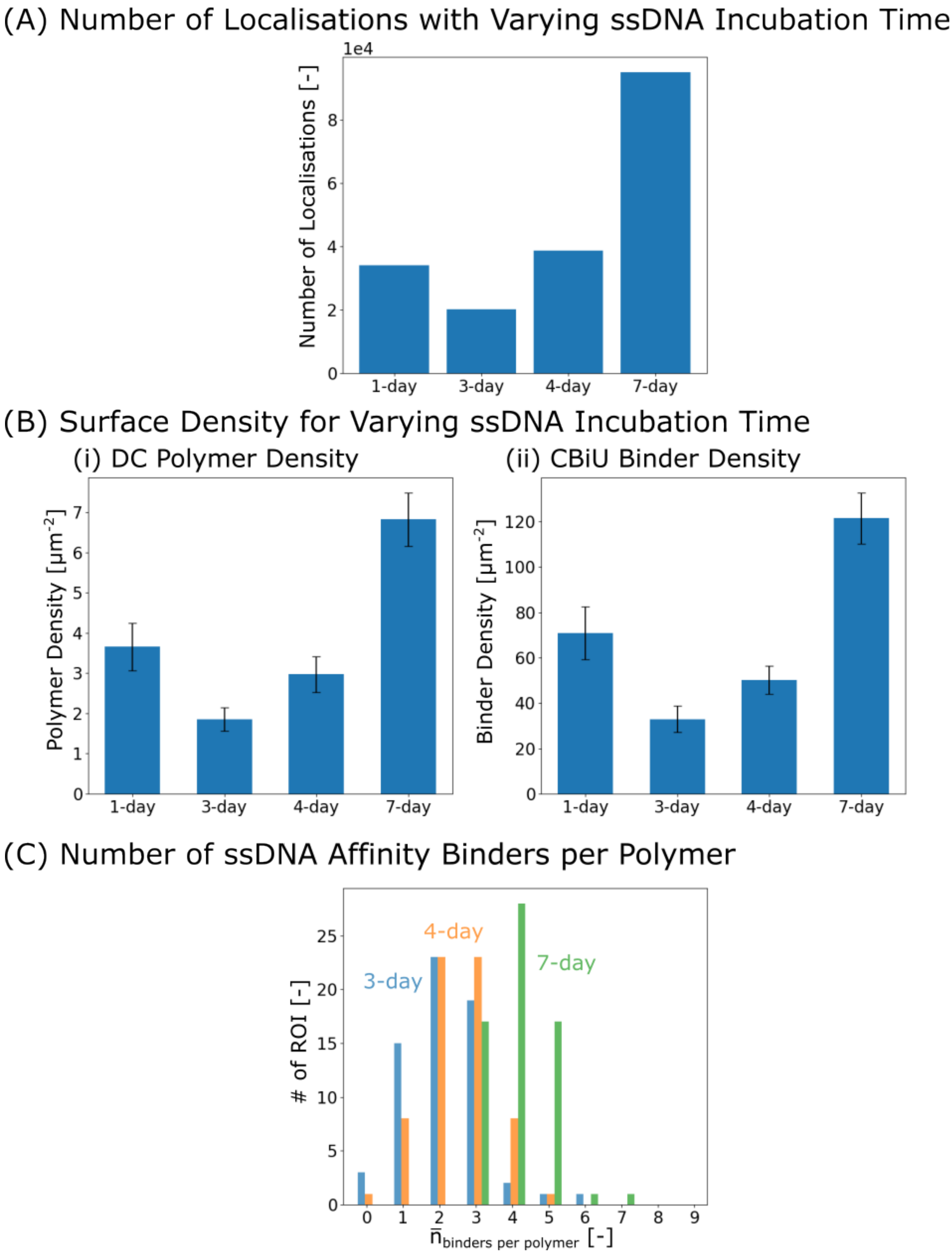

Figure 4: (A) The number of localisations shows an increase for increasing ssDNA incubation time. (B) The DC polymer density (i) and the CBiU binder density (ii) increase with increasing ssDNA incubation time. The error bars indicate the standard deviation between ROIs in an experimental data set. (C) Increasing ssDNA concentration incubation time increases the average number of conjugated ssDNA affinity binders per polymer with less variance.

3.4 Understanding the Combined Incubation System

The following experiments combined the PBS incubation with the ssDNA concentration incubation to understand the effects of both incubation steps. The polymer is first incubated in PBS for 4 days, followed by a 3-day incubation with various concentrations. The ssDNA incubation concentrations used are 0.5, 2, and 5 μM.

The control experiments in Figure 5A display a positive control (10 bp overlap between docking and imager strands) and a negative control (no complementarity between docking and imager strands). The positive control yielded a higher number of localisations compared to the negative control. For the 2 and 5 μM concentration, the negative control showed a similar number of localisations as the 4-day PBS incubation experiments, indicating that only background is detected. However, for the 0.5 μM concentration, the negative control showed a higher number of localisations compared to the positive control. A hypothesis for the high number of localisations is attributed to unconjugated azide-functionalised polymers that interact with the imager strands, similarly to a low incubation time. To validate this hypothesis, repeats are recommended to confirm the anomaly or show that this is normal behaviour for this concentration. As the localisation may originate from non-specific interactions, 0.5 μM concentration will be excluded for the spatial molecular analysis. Next, the surface densities, shown in Figure 5B, increase with increasing ssDNA incubation concentration. The ssDNA-functionalised polymer density in Figure 5B(i) increases from $(3.7 \pm 0.5)\mu m^{-2}$ to $(8.7 \pm 0.8)\mu m^{-2}$ with increasing ssDNA incubation concentration. The increase in ssDNA-functionalised polymer density increases roughly 2.4-fold with a 2.5-fold increase in ssDNA incubation concentration. The surface binder density shown in Figure 5B(ii) increases from $(81.7 \pm 20.3)\mu m^{-2}$ to $(162.9 \pm 12.2)\mu m^{-2}$ with increasing ssDNA incubation concentration. The increase in surface binder density is not linearly correlated to the increase in ssDNA incubation concentration. The observed surface binder density has an increase of roughly 2-fold, while the increase in ssDNA incubation concentration is 2.5-fold. Furthermore, the average number of affinity binders per polymer has been determined, as shown in Figure 5C. There is no correlation shown where increasing the ssDNA concentration increases the average number of affinity binders per polymer. 2 μM showed an average number of (5.5 ± 1.7) ssDNA binders per polymer, and 5 μM showed an average of (3.7 ± 1.1) ssDNA binders per polymer.

A hypothesis for the difference in the increase of the surface density is that the docking strands are less accessible for the imager strands. Which also explains the lower number of average affinity binders per polymer at 5 μM. Less accessible docking strands could result in a longer unbound state lifetime. An increase in the unbound state lifetime will decrease the association rate, which will also decrease the number of affinity binders per polymer (as shown in Equation 1). As the CLiU analysis does not compensate for the effect of inaccessibility, it can introduce artifacts in the quantification of high-density surfaces. DNA-origami structures can be used to determine the threshold of intermolecular distances where accessibility becomes an issue. Another hypothesis for the lower number of conjugated affinity binders per polymer is repulsion between the DBCO-ssDNA strands in solution. The distance between the DBCO-ssDNA strands increases at higher concentrations in solution, making conjugation near each other less likely. This will lead to a higher number of polymers with at least one docking strand, increasing the number of binders on the surface with a lower average binders per polymer.

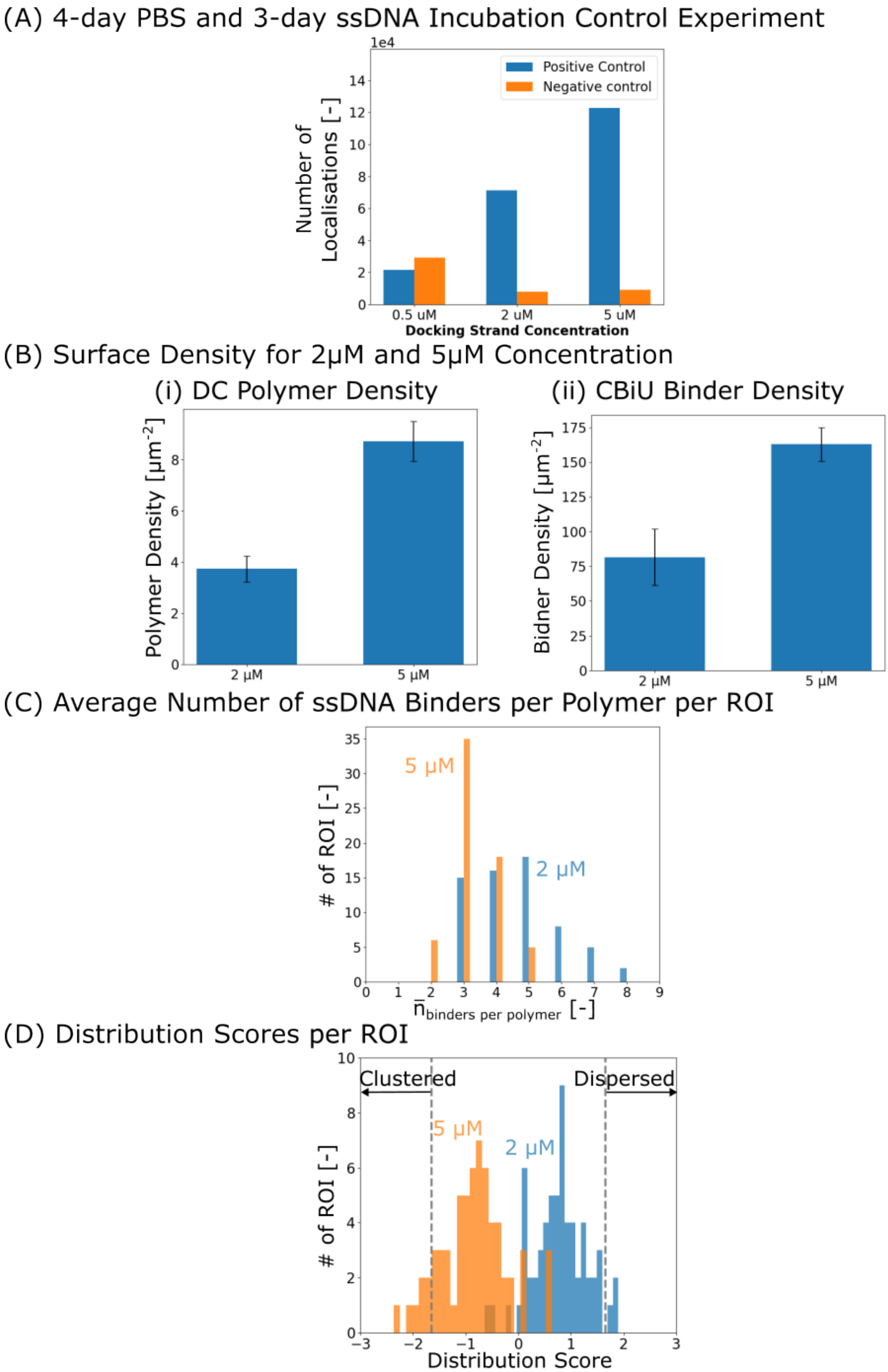

Figure 5: (A) The number of localisations per FOV with different concentrations of ssDNA solution, both positive and negative control. (B) The polymer density (i) and binder density (ii) are shown for 2 μM and 5 μM concentration. The error bar indicates the standard deviation per ROI. (C) The average number of affinity binders per polymer per ROI. (D) The spatial distribution with the 5% confidence (dotted lines) to visualise if the distribution is more clustered or dispersed per ROI.

3.5 Spatial Heterogeneity of Polymer

The spatial distribution was analysed with the data obtained from section 3.4. Figure 5D shows that the spatial distribution of the ssDNA-functionalised polymer is randomly distributed for both concentrations. Furthermore, the higher concentration shows a slight bias towards clustering compared to the lower concentration, which leans more towards a dispersed distribution. A Welch's t-test has been used to compare whether the two samples have an equal mean. The null-hypothesis is rejected (with a p-value of $4.0 \cdot 10^{-31}$), which means that there is a significant difference between the samples, showing that the incubation concentration influences the spatial distribution. This suggests a tendency towards a clustered distribution at higher incubation concentrations. The underlying source of this phenomenon remains unclear. Further investigation to understand this phenomenon is recommended. Still, the observed spatial distribution of both concentrations falls within the 5% confidence interval and shows a randomly distributed behaviour.

However, the reliability of the spatial distribution determination remains uncertain. The accuracy relies on several factors: the precision of the localisations, the sampling ratio, the surface density, and the underlying true spatial distribution. Inaccurate localisation can lead to misidentification of the polymer density, which will affect the nnd estimation. Additionally, if too few ssDNA-functionalised polymers are detected by the mean-shift algorithm, the surface might be undersampled, affecting the determination of the spatial distribution.

Nevertheless, the determined spatial distribution can be used as an indication of the underlying polymer distribution. There are several approaches to improve the determination of the spatial distribution. For example, a higher resolution microscopy technique, such as Resolution Enhancement by Sequential Imaging (RESI) [38], can improve the localisation precision. Additionally, optimising the protocol can improve the ssDNA-functionalised polymer density, which will in turn improve sampling of the surface. A machine learning tool could be implemented to estimate the distribution scores. The model can be trained to identify relations between the localisation precision, sampling ratio, surface densities, and the underlying true spatial distribution. By understanding the relation between the parameters, the model could determine the spatial distribution score.

4 Conclusions

In this study, surface densities and spatial distributions have been analysed for a high-density bio-functionalised PAcrAm™-g-PMOXA coated surface using DNA-PAINT. Three different analysis methods were used to determine surface densities, and one analysis method to determine the spatial distribution. The first method determines the ssDNA-functionalised polymer density, called Direct Counting (DC), via a mean-shift clustering algorithm. The following method determines the average number of affinity binders per polymer, called Kinetic Counting (KC). It uses the unbound state lifetimes of the binding events to determine the average number of affinity binders per polymer. Both methods are limited by the experimental variables. Monte-Carlo simulations were used to compensate for the experimental limitation. Compensation for Lifetime Undersampling (CLiU) was used to compensate for the overestimation via KC. Compensation for Binder Undersampling (CBiU), the third method, compensates for the number of binding events observed during the measurement to determine the affinity binder density. Lastly, the spatial distribution is determined with a Clark-Evans test to compare the ssDNA-functionalised polymer density with a complete spatial randomness hypothesis to determine if the distribution is clustered, random, or dispersed.

First, the effects of an additional PBS incubation step have been investigated. Increasing the duration of the incubation step decreases the number of non-specific interactions. The hypothesis for the high number of localisations for a short incubation time is a non-homogeneous polymer layer on the surface. The surface heterogeneity can be characterised with patches of low-fouling areas, where the surface is not fully covered with the polymer, there are unbound polymer tail-ends, or there are loops present within the polymer backbone. After a 4-day incubation period in PBS, the polymer showed a reproducible low number of localisations. This indicates that the low-fouling layer has been fully formed after a 4-day incubation period in PBS. This shows that the polymer is capable of rearranging to an equilibrium configuration, which suggests two possibilities. First, the low-fouling layer becomes stable after the formation of the covalent bonds. Second, sufficient covalent bonds are formed after 4 days of PBS incubation to form a homogeneous, low-fouling layer.

Next, the effect of varying the incubation time with the DBCO-ssDNA solution has been investigated. An increase in the ssDNA-functionalised polymer density and the affinity binder density has been observed. Additionally, an increase in the average number of affinity binders per polymer is also observed with a smaller variance. This indicates that increasing the incubation time with the DBCO-ssDNA solution also increases the number of conjugated binders onto the polymer with a smaller variance between each polymer. However, 1-day incubation with the solution did show a higher number of localisations compared to the 3-day incubation. A hypothesis for this deviation of the trend stems from the non-specific interactions due to the heterogeneity of the low-fouling layer, similar to the 1-day incubation with PBS. Another possibility is interaction with unconjugated azides due to the short incubation time. This results in non-specific interactions between the unconjugated azides and the imager strands.

Following this, the PBS incubation and ssDNA incubation were combined to see the effect of the combined incubation system. For these experiments, a 4-day

PBS incubation was used, followed by a 3-day incubation in different DBCO-ssDNA incubation concentrations. The negative control showed a low number of localisations that are comparable with the number of localisations detected for the 4-day incubation in PBS, except the 0.5 μM ssDNA incubation concentration. A hypothesis for the high number of localisations is non-specific interactions caused by unconjugated azides due to the low incubation concentration. To validate this hypothesis, further investigation is recommended to confirm the anomaly. Furthermore, the ssDNA-functionalised polymer density and the affinity binder density increased with increasing ssDNA concentration. This shows that more polymers are conjugated with at least one ssDNA docking strand. However, the observed increase of the ssDNA-functionalised polymer and the affinity binder density did not increase equally. This is also reflected in the observed average number of affinity binders per polymer, where it decreased with increasing concentration. Two hypotheses are proposed to address this problem. First, a higher concentration could make the docking strand less accessible. This could result in a longer unbound state lifetime, reducing the quantified number of affinity binders per polymer. Second, a stronger repulsion between the DBCO-ssDNA strands in solution. This can introduce difficulties in conjugation near each other, resulting in a higher increase in ssDNA-functionalised polymer density compared to the increase of the affinity binder density.

Lastly, the spatial distribution of the ssDNA-functionalised polymer was tested against the CSR hypothesis using the Clark-Evans test. The results showed a random distribution for both 2 and 5 μM DBCO-ssDNA incubation concentration. The higher concentration showed a bias towards a more clustered distribution. The underlying source remains unclear; further investigation to understand this phenomenon is recommended. Nevertheless, the spatial distribution of both concentrations shows a random distribution. The reliability of the spatial distribution remains uncertain due to several factors, such as the precision of the localisations and the sampling ratio. These results show an indication of what the surface distribution is. To improve the interpretation of the observed distribution score, a higher accuracy could be obtained with either a higher resolution microscopy technique, optimising the protocol for increased sampling, or implementing a machine learning tool to estimate the distribution score based on experimentally obtained sampling ratios.

The goal of this study is to characterise the spatial heterogeneity of biomolecule functionalised PAcrAm™-g-PMOXA coated surface using DNA-PAINT. The results provide a fundamental concept of ssDNA-functionalised PAcrAm™-g-PMOXA coated surface and show promising outcomes. It is recommended to repeat some of the experiments for a more accurate interpretation of the results and to test the given hypotheses. Furthermore, the scope of this study is to characterise the biomolecule functionalised PAcrAm™-g-PMOXA coated surface and not optimisation. The insights of this study can guide the optimisation process for PAcrAm™-based bio-functionalised surfaces, aiming for an optimal balance of incubation time and observed surface densities. The interest in a new low-fouling polymer increased due to the limitations of PLL-g-PEG in biomedical applications. A direct comparison study between PLL-g-PEG and PAcrAm™-g-PMOXA as a low-fouling surface coating in a biosensor should be conducted. Various properties of PAcrAm™-g-PMOXA have been studied and analysed, but the characterisation does not directly predict the polymer behaviour used in application. Lastly, it would be interesting to research what influences the low-fouling layer formation. This study showed a decrease in non-specific interactions with increasing incubation time in PBS. Understanding the molecular mechanism of forming the low-fouling layer could provide insights into the underlying mechanics for covalent bond formation and the dependency of the low-fouling layer on the covalent bonds.

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Appendix A Polymer Surface Estimation

To have an estimate about the spatial distribution of PAcrAm™-g-PMOXA(-N_3), a few theoretical calculations have been made in advance to get an expectation value of what the numbers might be.

The backbone length of PAcrAm™ is calculated by using the bond length between two carbon atoms ($l_{C-C} = 0.154$ nm), the bond angle angle between two carbon atoms ($\theta_{C-C} = 120°$), the number carbon-carbon bonds in one PAcrAm™ monomer ($n_{C-C} = 2$), and the number of monomers in the polymer ($n_{AcrAm} = 95$). The backbone length can be calculated using Equation A, with the assumption that the polymer is well extended on the surface.

$$l_{backbone} = n_{AcrAm} \cdot n_{C-C} \left(l_{C-C} sin \left(\frac{\theta_{C-C}}{2} \right) \right) \tag{A}$$

The mixing ratio of 1% v/v does not explain the number of azide-functionalised polymers on the surface. To determine the amount, the mixing ratio needs to be derived to the number ratio. The mixing ratio is given by Equation B where $c^{mass}_{tot} = c^{mass}_{PaS-N3} + c^{mass}_{PaS}$.

$$r_m = \frac{c^{mass}_{PaS-N3}}{c^{mass}_{tot}} \tag{B}$$

The number ratio is given by Equation C where $n_{PaS,tot} = n_{PaS} + n_{PaS-N3}$.

$$r_n = \frac{n_{PaS-N3}}{n_{PaS,tot}} \tag{C}$$

Here, c^{mass}_x denotes the mass concentration [mg/mL] and n_x the number of molecules of polymer x. The correlation between the mixing ratio and the number ratio is shown in Equation D.

$$\begin{aligned}
r_n &= \frac{n_{PaS-N3}}{n_{PaS,tot}} \\
r_n &= \frac{n_{PaS-N3}}{n_{PaS} + n_{PaS-N3}} \\
\text{Where } n &= c^{mol}_x N_A V \\
r_n &= \frac{c^{mol}_{PaS-N3}}{c^{mol}_{PaS} + c^{mol}_{PaS-N3}} \\
r_n &= \frac{\frac{c^{mass}_{PaS-N3}}{MW_{PaS-N3}}}{\frac{c^{mass}_{PaS}}{MW_{PaS}} + \frac{c^{mass}_{PaS-N3}}{MW_{PaS-N3}}} \\
r_n &= \frac{r_m c^{mass}_{tot}}{(1 - r_m) c^{mass}_{tot} \frac{MW_{PaS-N3}}{MW_{PaS}} + r_m c^{mass}_{tot}}
\end{aligned} \tag{D}$$

Where c^{mol}_x denotes the molar concentration [mol/mL], MW_x the molecular weight [g/mol] of polymer x, N_A is the Avogadro constant, and V the volume. As N_A and V are constant, n is proportionate to c^{mol}_x.

The polymer and azide-functionalised densities can be calculated using Equation E.

$$
\begin{aligned}
\sigma_{tot} &= \frac{1}{(1-r_n)A_{PaS} + r_n A_{PaS-N3}} \\
\sigma_{PaS-N3} &= \frac{r_n}{(1-r_n)A_{PaS} + r_n A_{PaS-N3}}
\end{aligned} \tag{E}
$$

Where σ_{tot} is the total polymer surface density $[\mu m^{-2}]$, σ_{PaS-N3} the azide-functionilised polymer density $[\mu m^{-2}]$, and A_x the area $[\mu m^2]$ of polymer x.
The area of the polymer is calculated by multiplying the length of one polymer by 2 nm, which is the interchain distance of a polyelectrolyte polymer [39]. The calculated values are shown in Table A with a mixing ratio of 0.01, a total mass concentration 0.1 mg/mL, and 95 numbers of monomers in one PAcrAm™-g-PMOXA polymer. The range given by SuSoS is 80 to 110 monomers per PAcrAm™-g-PMOXA polymer. As these are preliminary calculations, no further investigation on the actual number of monomers has been conducted.

Table A: Parameters of PAcrAm™-g-PMOXA(-N3)

MW AcrAm monomer	[g/mol]	71.08
Number of monomers	[-]	95
Molecular Weight of backbone	[g/mol]	6753
Backbone length	[nm]	23.81
MW Silane side chain	[g/mol]	161.32
Grafting ratio Silane side chain	[-]	2.5
Average number of Silane side chains	[-]	38
MW Amine side chain	[g/mol]	116.20
Grafting ratio Amine side chain	[-]	2.5
Average number of Amine side chains	[-]	38
MW PMOXA(-N3) side chain	[g/mol]	4048 (6100)
Grafting ratio PMOXA(-N3) side chain	[-]	5
Average number of PMOXA(-N3) side chains	[-]	19
MW_{total}	[g/mol]	94,210 (133,198)
$A_{PaS(-N3)}$	[nm^2]	47.64
σ_{PaS-N3}	$[\mu m^{-2}]$	149.91
σ_{ssDNA}	$[\mu m^{-2}]$	1340.18

There are 19 azides available on one polymer to conjugate with ssDNA strands. The distance between each azide-functionalised PMOXA strand is 1.2 nm. This is smaller than the size of a partially double-stranded DNA, which is roughly 2 nm. Thus, not all azides can be conjugated and an estimation of 9 ssDNA can be conjugated onto one polymer. DNA-PAINT can be used if the distance between two polymers are larger than the resolution of DNA-PAINT, which is roughly 8 nm. Otherwise, the PSFs of the polymers can overlap resulting in unfavourable data for analysis. The distance between two clouds can be calculated using Equation F.

$$
d_{PaS-N3} = \frac{1}{2\sqrt{\sigma_{PaS-N3}}} \tag{F}
$$

This results in a distance of roughly 41 nm, making it suitable for DNA-PAINT

Appendix B Repeat 1-day ssDNA Incubation

As mentioned in section 3.3, repeats of the 1-day incubation with ssDNA solution showed similar behaviour between different samples. This explains that the observed behaviour is not an anomaly. The results are shown in Figure A. The number of localisations, shown in Figure A, does not deviate far from each other. Comparing with the number of localisations obtained with the additional PBS incubation step (section 3.2), the number of localisations are similar to the 1-day incubation in PBS and the repeats of no PBS incubation. This comparison could confirm that the detected signal include non-specific interactions. Thus, the localisations cannot be interpreted as the ssDNA-functionalised polymer.

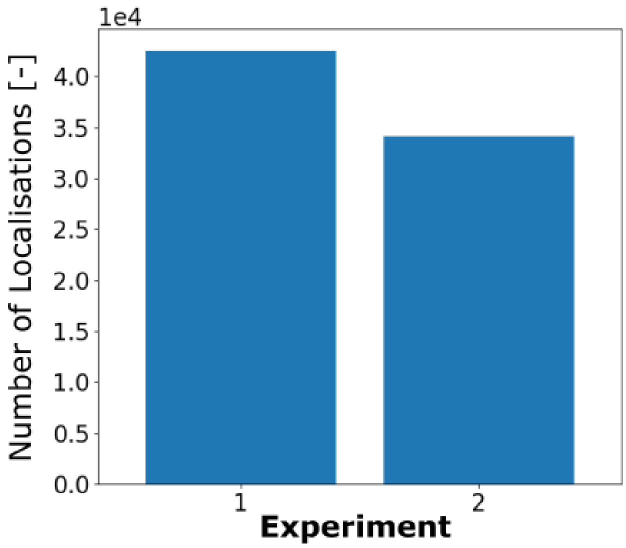

Figure A: Two different experiments of the 1-day incubation in ssDNA solution with a similar number of localisations.